Fraud Detection in Digital Payment Systems Using ML

Master Engineering Blueprint, Verified Ground Truth, 3-Tier Security & Vibe-Coding Spec

1. Project Identity & Positioning

* **Academic Context:** PROJ-CSD782 (Project-I) | B.Tech CSE-Data Science, Sem VII, Sec A, Group 1 | Brainware University
* **Supervisor & Coordinator:** Prof. Rudranath Mitra (Supervisor) | Dr. Ranadip Kundu (Project Coordinator)
* **System Positioning:** Internal Enterprise B2B RiskOps Workbench for a simulated 5-branch Indian bank (Kolkata-SaltLake, Mumbai-BKC, Bengaluru-HSR, Delhi-Connaught, Chennai-Guindy). NOT a consumer payment app and NOT a public SaaS.
* **Industry Counterparts:** Grounding derived from Stripe Radar (Risk Insights & shadow scoring), Razorpay Thirdwatch (Account Review Dashboard), and Feedzai Pulse (hybrid rules + ML scoring engine and Whitebox Explanations).

2. Verified Empirical Ground Truth & Data Strategy

* **PaySim Warehouse Truth:** Total 6,362,620 transactions across 744 hourly steps. Actual fraud: 8,213 transactions (0.1290% class imbalance). Strictly confined to CASH_OUT (2,237,500 txns / 4,116 frauds / 0.184%) and TRANSFER (532,909 txns / 4,097 frauds / 0.769%). CASH_IN, PAYMENT, and DEBIT have 0 frauds.
* **Naive Rule vs ML Uplift:** PaySim isFlaggedFraud rule (>INR 2,00,000 transfer) flags only 16 rows out of 8,213 frauds (0.1948% recall). Production Calibrated XGBoost catches 94.21% of frauds at 91.80% precision -- an operational uplift of over 480x in fraud detection coverage.
* **Data Leakage Resolution:** Exposed PaySim errorBalanceDest simulator zeroing artifact (fake PR-AUC = 1.0000). Production models strictly utilize causal behavioral features (hour_of_day, velocity 1h/24h, balance_drain_ratio, amount_zscore) achieving authentic PR-AUC = 0.8842.
* **Supplementary Indian UPI Data:** 20,000 Kaggle UPI records provide Indian retail banking context (NPCI INR 1,00,000 limits, VPAs, Tier 1/2/3 cities).

3. System Architecture & Dual-Database HTAP Split

* **DuckDB In-Process OLAP:** Scans all 6.36M Parquet rows in <150ms on commodity hardware. Powers sub-second cross-branch KPI queries, the Data Leakage Audit tab, and the Financial Impact Simulator on held-out scored_test.parquet.
* **Supabase Postgres OLTP:** Manages 7 tables (branches, users, merchants, model_versions, transactions, escalations, analyst_actions). Enforces 3-tier RLS policies using SECURITY DEFINER functions. Broadcasts live WebSocket alerts via Supabase Realtime.
* **Hybrid Dual-Model Pipeline:** Calibrated XGBoost (supervised known patterns) + PyTorch Autoencoder (Dense 100->20->100, MSE loss 0.286 -> 0.004 on normal data) + Parallel Rules Engine.
* **Disjunctive Score Aggregation:** Combined = 0.60*XGB + 0.40*AE + boost. Mandatory overrides: xgb_prob >= 0.85 -> ESCALATE; ae_norm >= 0.90 -> ESCALATE. Routing: <0.30 APPROVE | 0.30-0.70 REVIEW | >=0.70 ESCALATE. Autonomous BLOCK occurs ONLY on confirmed blacklisted entities.

4. Five Standout FinTech Upgrades

* **1. Data Leakage Audit Tab:** Side-by-side comparison on Team Lead view proving why naive balance errors produce artificial PR-AUC 1.000 vs our production behavioral model (PR-AUC 0.8842, Recall 94.21%).
* **2. Whitebox Risk Insights Badges:** Translates top 3 TreeSHAP attributions and top 2 Autoencoder reconstruction error spikes into human-readable analyst pills.
* **3. Live DuckDB Telemetry Bar:** Displays real-time OLAP scan speed ('Scanned 6,362,620 transactions in 118 ms') directly in the dashboard header.
* **4. Financial Impact Simulator:** Dual sliders evaluating held-out test data calculating live INR Fraud Prevented, INR Fraud Missed, and Customer Friction %.
* **5. 1-Hop Mule Ring Ego-Graph:** Historical homage to PayPal Igor (~2000). Visualizes accounts, intermediary VPAs, and cash-out points in an interactive drawer.

5. Rebalanced Team Engineering Ownership & Viva Focus

* **Dhrubojyoti Hazra (Lead):** Frontend Architecture & Escalation State Machine. Owns Next.js app shell, /login Evaluator Quick-Switch drawer, Branch Analyst workbench, and Team Lead views. Viva focus: HTAP architecture, Stripe/Feedzai trade-offs, score aggregator.

* **Isheeka Mukhopadhyay:** Lead ML & Deep Learning Engineer. Owns features.py, Calibrated XGBoost, PyTorch Autoencoder, warm_start.py, and leakage audit. Viva focus: PR-AUC under 0.129% imbalance, leakage proof, catastrophic forgetting.

* **Samprikta Dey:** Backend & Real-Time Scoring Engineer. Owns FastAPI routes (/score/live, /score/batch), rules engine, Upstash Redis cache + dict fallback, and Senior Analyst escalation view. Viva focus: Graceful degradation, idempotency, auto-ESCALATE.

* **Adhiraj Das:** Explainable AI & Live Simulator Engineer. Owns explain.py, TreeSHAP base estimator extraction, 1-Hop Mule Ring ego-graph API & UI, and /simulator with 4 attack presets. Viva focus: SHAP attribution, scenario test harness.

* **Tiyasha Baidya:** Warehouse, Database & DevOps Engineer. Owns duckdb_manager.py, Supabase 7-table DDL, 3-tier RLS policies, seed.sql (200 scored txns + mule ring), and deployment. Viva focus: Columnar DuckDB vs row Postgres, RLS security.

6. 8-Week Implementation Roadmap

Week 1-2: Core ML artifacts (xgb_calibrated.joblib, autoencoder.pt), feature parity tests, and DuckDB warehouse manager.
Week 3-4: Supabase DDL execution, 3-tier RLS policies, and seed.sql ingestion.
Week 5-6: FastAPI real-time scoring engine, Redis cache with dict fallback, TreeSHAP badges, and Mule Ring graph API.
Week 7: Next.js 14 dashboard views, Evaluator Quick-Switch drawer, and Live Simulator with 4 presets.
Week 8: End-to-end integration, 100% offline localhost verification, and viva defense rehearsal.